

Web Programming

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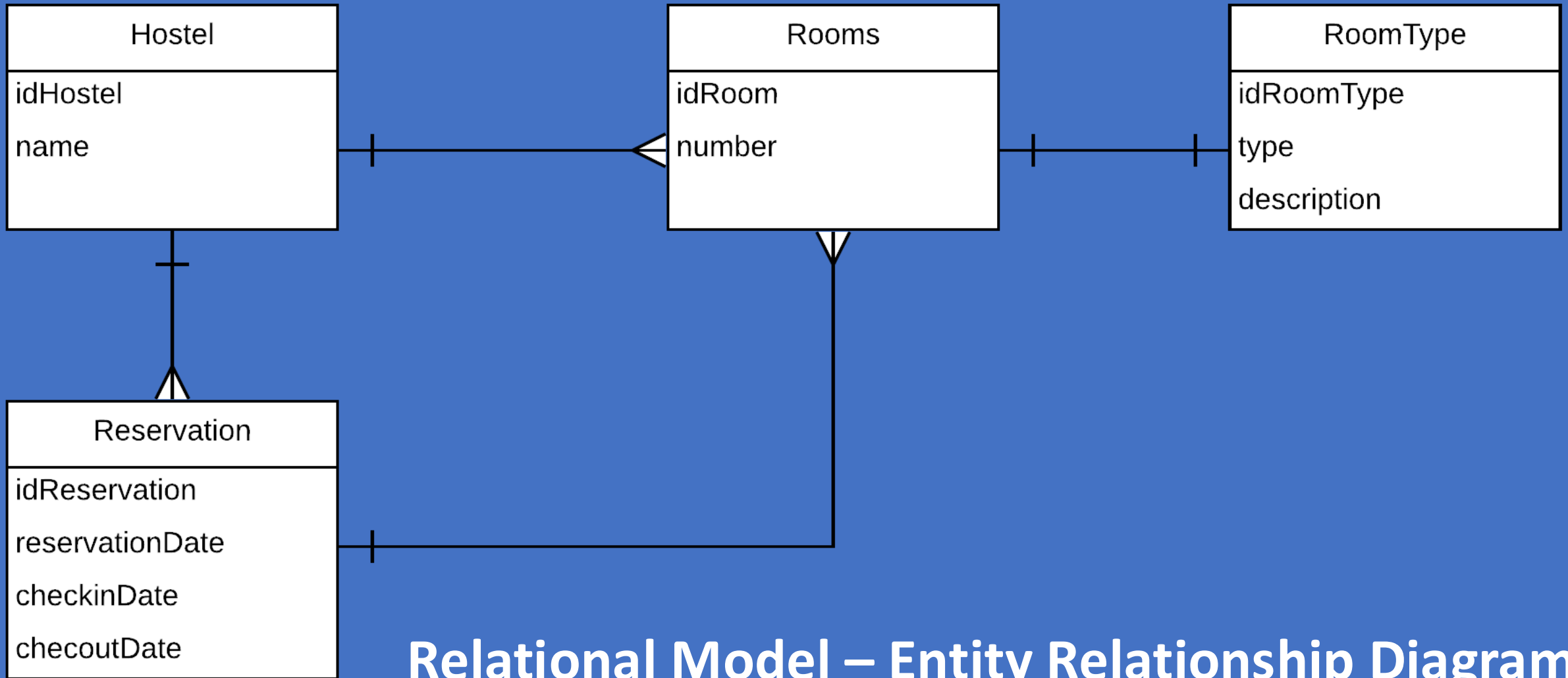
Web programming foundations

**Back-end languages, CMSs,
APIs and services**

Database management systems

Lecture: Database management systems

Module: Back-end languages, CMSs, APIs & Services



Relational Model – Entity Relationship Diagram

SQL Databases



- **Web developers, especially back-end specialists, need to be able to work with databases.**
 - ✓ Developers need to know database languages like Structured Query Language (SQL).
- **SQL is a popular way to store and retrieve information.**



SQL Databases



- You may also hear **SQL** pronounced as **sequel**.
- **SQL databases** have clearly **defined structures** which is sometimes referred to as the **schema**.



NoSQL Databases

- There's **another** important **category** of **databases** that you **should** be **familiar** with.
 - ✓ NoSQL databases.
- NoSQL databases have no pre-defined structure and store information in dynamic schemas.



NoSQL Databases

- **NoSQL databases** can actually be **faster** and **more efficient** because some of that **structure** in **SQL** ends up as **overhead** and **makes** them **run slower**.



Back-end languages and frameworks

Lecture: Back-end languages and frameworks

Module: Back-end languages, CMSs, APIs & Services

Back-end languages

- In addition to databases, we need a language/framework that can take the data from databases and merge it with templates to display customized info.
 - ✓ Dynamic language/framework.

Back-end languages and frameworks

- **Examples of languages/frameworks that combine templates with data and build custom pages for each user:**
 - ✓ Java/Jakarta EE/Spring/Spring Boot
 - ✓ PHP/Laravel
 - ✓ JavaScript/Node.js/Express.js
 - ✓ C#/.NET
 - ✓ Ruby/Ruby on Rails
 - ✓ Python/Django/Flask

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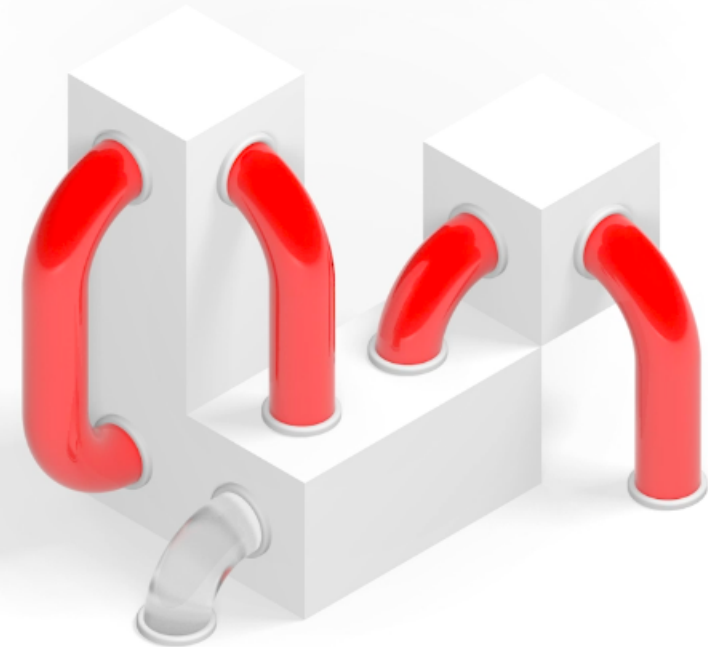
The PHP Framework for Web Artisans

Laravel is a web application framework with expressive, elegant syntax. We've already laid the foundation — freeing you to create without sweating the small things.

Documentation



Watch Laracasts



<https://laravel.com>

Express 4.17.1

Fast, unopinionated, minimalist web
framework for **Node.js**

```
$ npm install express --save
```

 **Express 5.0 alpha documentation is now available.**

The alpha [API documentation](#) is a work in progress. For information on what's in the release, see the Express [release history](#).

Web Applications

Express is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.

APIs

With a myriad of HTTP utility methods and middleware at your disposal, creating a robust API is quick and easy.

Performance

Express provides a thin layer of fundamental web application features, without obscuring Node.js features that you know and love.

Frameworks

Many [popular frameworks](#) are based on Express.

<https://expressjs.com>

.NET

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Supported on macOS, Windows, and Linux



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Build web apps and services for macOS, Windows, Linux, and Docker.



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Use a single codebase to build native mobile apps for iOS, Android, and Windows.



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Create independently deployable microservices that run on Docker containers.



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Consume existing cloud services, or create and deploy your own.



Machine Learning

Add vision algorithms, speech processing, predictive models, and more to your apps.



Game Development



Internet of Things

Feedback

<https://dotnet.microsoft.com>



Imagine what you could build if you learned Ruby on Rails...

Learning to build a modern web application is daunting. Ruby on Rails makes it much easier and more fun. It includes **everything you need** to build fantastic applications, and **you can learn it** with the support of our **large, friendly community**.



Latest version — Rails 6.1.3.2 released May 5, 2021



<https://rubyonrails.org>

Django makes it easier to build better Web
apps more quickly and with less code.

[Get started with Django](#)

Meet Django

Django is a high-level Python Web framework that encourages rapid development and clean, pragmatic design. Built by experienced developers, it takes care of much of the hassle of Web development, so you can focus on writing your app without needing to reinvent the wheel. It's free and open source.



Ridiculously fast.

Django was designed to help developers take applications from concept to completion as quickly as possible.

[Download latest release: 3.2.4](#)[DJANGO DOCUMENTATION >](#)

Support Django!



Harry Percival donated to the Django
Software Foundation to support Django
development. Donate today!

<https://www.djangoproject.com>

Django security releases issued: 3.2.4, 3.1.12, and
2.2.24

Content Management Systems (CMS)

Lecture: Content Management Systems (CMS)

Module: Back-end languages, CMSs, APIs & Services

Content Management Systems - CMS

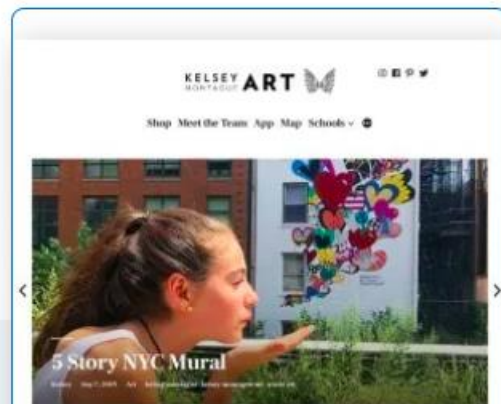
- Examples of CMSs that combine templates with data and build custom pages for each user.
 - ✓ WordPress
 - ✓ Drupal
 - ✓ Joomla



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<https://www.drupal.org>



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APIs & services

Lecture: APIs and services

Module: Back-end languages, CMSs, APIs & Services

APIs & Services

- A **developer doesn't** always **build** an **entire project**.
 - ✓ He/she often **uses** or **creates** a **number** of **services** and **APIs**.
- A **service** is a **feature offered** by a **server** that **provides access** to **some information**.
 - ✓ **Developers** often **create** these to **simplify** the **sharing** of **data**.

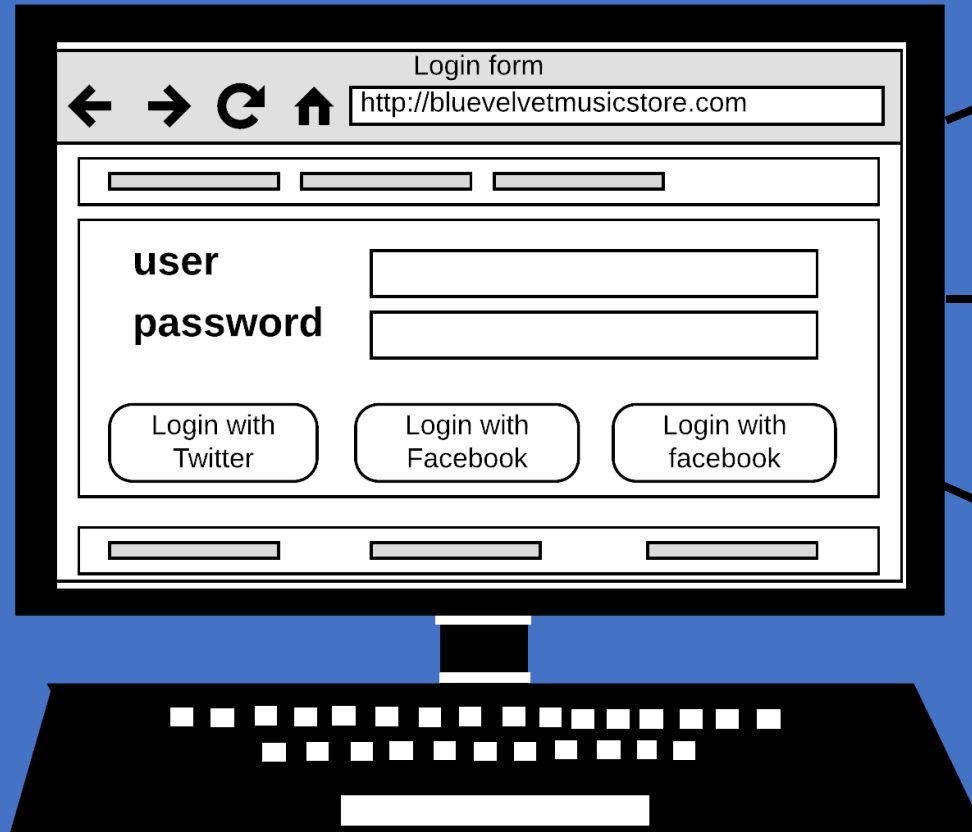
APIs & Services

- A service, allows for instance, **some sites to log in through your Facebook, Google, or Twitter accounts.**
 - ✓ This is made possible by something called an **API**, or **Application Programming Interface.**
- An **API** is a **small application** used to **formalize the way** that the **information is requested.**

APIs & Services

- Most **APIs** are considered **RESTful**, which is a **service based** on **HTTP**.
- **RESTful services** use **HTTP verbs** like **GET** and **POST** to **retrieve** and **send information** to **manage data**.
 - ✓ The **requests** are **usually returned** in a **format called JSON**, or **JavaScript Object Notation**.
- **JSON** is a **text-based format** that **easily converts** into **JavaScript objects**.

API & services



Login API



Login API



Login API



RESTful Web Services

- As a **developer**, you'll need to be **comfortable** with **using** and **managing data** from a **service** like **REST**.
- **Back-end developers** create their own **web services** and **APIs**.

APIs & Services

- Learning to work with APIs and services is a critical skill for developers.
- How much you need to learn depends on the type of developer you want to be.



Featured: [Kubernetes to the Cloud with Spring Boot and JHipster](#)

OAuth 2.0

OAuth 2.0 is the industry-standard protocol for authorization. OAuth 2.0 focuses on client developer simplicity while providing specific authorization flows for web applications, desktop applications, mobile phones, and living room devices. This specification and its extensions are being developed within the [IETF OAuth Working Group](#).



Questions, suggestions and protocol changes should be discussed on the [mailing list](#).

New Video Course!



OAuth 2.0 assures request comes from an authorized source

The Nuts and Bolts of OAuth 2.0

by Aaron...

<https://oauth.net/2>

OAuth 2.0

- **OAuth 2.0 can act like an intermediary so that requests can be securely managed without having to constantly share passwords between two systems.**
 - ✓ That's **why** you can tell an **application** like **WordPress** to post to your **Twitter** feed.
 - ✓ Once you've **authorized** your **WordPress** account to **post** to **Twitter**, the **two** **services** **share** a **secret token**, and that **makes communication easier**.

Questions?